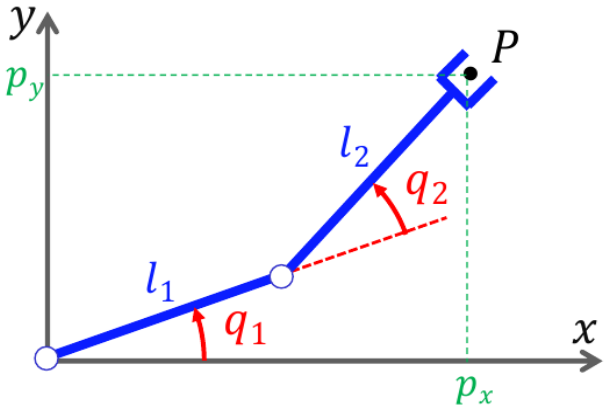
Introduction to Robotics, Lecture 5

Workspace and Singularities

MIPT

Center for Cognitive Modeling

Which positions can point *P* reach?



Two links: l_1, l_2.

Joint angles: q_1, q_2.

P is the selected end-effector point.

Notation. l_i: link length; q_i: joint angle; *P*: task point.

If *P* is reachable, can it move in every direction?

Position workspace of point P

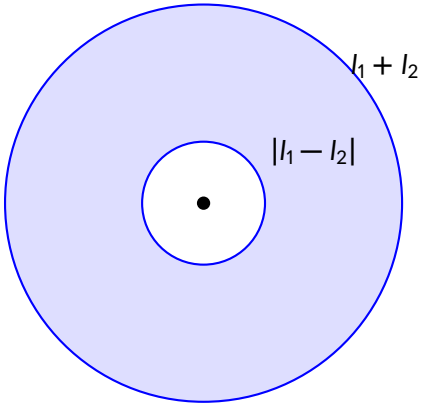
$$r^2 = l_1^2 + l_2^2 + 2 l_1 l_2 \cos q_2$$

Since $-1 \leq \cos q_2 \leq 1$,

$$|l_1 - l_2| \leq r \leq l_1 + l_2.$$

If $l_1 = l_2$, the inner radius is zero.

Notation. r: distance from the base to P.



Question: On the outer boundary, can P move radially outward instantaneously?

The different types of workspaces

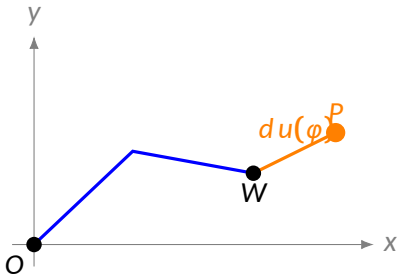
- *constant orientation workspace* or *translation workspace*: all possible locations of the operating point *C* of the robot that can be reached with a given orientation
- *orientation workspace*: all the possible orientations that can be reached while *C* is in a fixed location
- *maximal workspace* or *reachable workspace*: all the locations of *C* that may be reached with at least one orientation of the platform.
- *inclusive orientation workspace*: all the locations of *C* that may be reached with at least one orientation among a set defined by ranges on the orientation angles. The maximal workspace is a particular case of inclusive orientation workspace for which the ranges for the orientation angles are $[0, 2\pi]$
- *total orientation workspace*: all the locations of *C* that may be reached with all the orientations among a set defined by ranges on the orientation Angles

The different types of workspaces

- *dextrous workspace*: all the locations of C for which all orientations are possible. The dextrous workspace is a particular case of total orientation workspace, the ranges for the rotation angles being $[0, 2\pi]$.
- *reduced total orientation workspace*: all the locations of C that may be reached with a subset of the orientation angles that may have any value in defined ranges, while the others may have arbitrary values. Such a workspace may be important for applications that do not involve all the d.o.f. of the robot. For example, if a 6 d.o.f. robot is used as a 5-axis machine-tool, the φ orientation parameter is not important
- *controllably dexterous workspace*: the subset of the dextrous workspace which does not contain any singular configuration

Why are these workspaces different?

Planar 3R arm: two 1 m links and a tool of length $d = 0.4$ m



$$W = P - d\,u(\varphi), \qquad \|W\| \le 2.$$

Fixed orientation

$\|P - d\,u(\varphi)\| \le 2$: a shifted radius-2 m circle.

At least one φ exists

$\min_\varphi \|P - d\,u(\varphi)\| = |\|P\| - d|$; maximum: $\|P\| \le 2.4$ m.

Every φ works

$\max_\varphi \|P - d\,u(\varphi)\| = \|P\| + d$; dextrous: $\|P\| \le 1.6$ m.

Notation. O: base; W: wrist; d: tool length; $u(\varphi)$: unit direction.

Computing a Workspace: FK and IK

The first choice is which direction to solve the kinematic map

Sample joint configurations (FK)	Test task-space poses (IK)
Compute $p = f(q)$ for many joint samples and mark the resulting poses.	Select poses p and ask whether an allowed q solves $f(q) = p$.

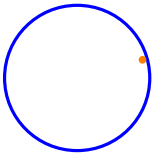
- Simple; no inverse kinematics is needed.
- Samples cover task space unevenly and may miss a boundary.

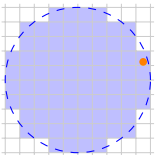
- Tests chosen poses and can count solution branches.
- Needs complete, reliable IK; numerical failure alone does not prove a pose unreachable.

Describing a Workspace: Geometry and Grid

The second choice is how to describe or test the reachable set

Geometric computation	Grid computation
$\|\|P - d\,u(\varphi)\|\| \leq 2$ gives an exact circle boundary for this simple arm; complex limits make analytic geometry harder.	Test regular nodes, using a geometric condition or IK. Limits and branch counts fit naturally, but the boundary is approximate.





Geometry checks the answer; a grid makes broader cases computable. In D dimensions, node count is of order $(L/h)^D$.

Notation. L: span; h: spacing; D: dimension.

Explore workspaces in the model

One geometry, different orientation requirements for point P

1. Set $P = (2.2, 0)$ m. Is it reachable for every tool orientation?
2. Compare **at least one** with **every** orientation in $[-60°, 60°]$.
3. Switch one workspace between geometric and grid computation. What changes?

Open the interactive model

http://localhost:8765/workspace.html

The maximum workspace $\supseteq$ the dextrous workspace. A reachable position still does not guarantee the required instantaneous velocity.

Primer on linear algebra

given a matrix J: $m \times n$ (m rows, n columns)

- rank $\rho(J)$ = max # of rows or columns that are linearly independent
  - $\rho(J) \leq \min(m, n) \Leftarrow$ if equality holds, J has full rank
  - if $m = n$ and J has full rank, J is nonsingular and the inverse J^{-1} exists
  - $\rho(J)$ = dimension of the largest nonsingular square submatrix of J
- range space $\mathcal{R}(J)$ = subspace of all linear combinations of the columns of J

  $$\mathcal{R}(J) = \{v \in \mathbb{R}^m : \exists \xi \in \mathbb{R}^n, v = J\xi\}$$ ← also called image of J
  - $\dim(\mathcal{R}(J)) = \rho(J)$
- null space $\mathcal{N}(J)$ = subspace of all vectors that are zeroed by matrix J

  $$\mathcal{N}(J) = \{\xi \in \mathbb{R}^n : J\xi = 0 \in \mathbb{R}^m\}$$ ← also called kernel of J
  - $\dim(\mathcal{N}(J)) = n - \rho(J)$
- $\mathcal{R}(J) \oplus \mathcal{N}(J^T) = \mathbb{R}^m$ and $\mathcal{R}(J^T) \oplus \mathcal{N}(J) = \mathbb{R}^n$ (direct sum of subspaces)
  - any element $v \in V = V_1 + V_2$ can be written as $v = v_1 + v_2$, $v_1 \in V_1, v_2 \in V_2$
  - … in a unique way if and only if $V_1 \cap V_2 = \{0\}$ (a 'direct' sum, not just a sum!)

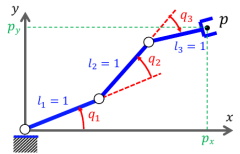
Mobility analysis in the task space

rank of a matrix

- $\rho(J) = \rho(J(q))$, $\mathcal{R}(J) = \mathcal{R}(J(q))$, $\mathcal{N}(J^T) = \mathcal{N}(J^T(q))$, etc. are locally defined, i.e., they depend on the current configuration q
- $\mathcal{R}(J(q))$ is the subspace of all "generalized" velocities (with linear and/or angular components) that can be instantaneously realized by the robot end-effector when varying the joint velocities $\dot{q}$ at the current q
- if $\rho(J(q)) = m$ at q ($J(q)$ has max rank, with $m \leq n$), the end-effector can be moved in any direction of the task space $\mathbb{R}^m$
- if $\rho(J(q)) < m$, there are directions in $\mathbb{R}^m$ in which the end-effector cannot move (at least, not instantaneously!)
  - these directions $\in \mathcal{N}(J^T(q))$, the complement of $\mathcal{R}(J(q))$ to task space $\mathbb{R}^m$, which is of dimension $m - \rho(J(q))$
- if $\mathcal{N}(J(q)) \neq \{0\}$, there are non-zero joint velocities $\dot{q}$ that produce zero end-effector velocity ("self motions")
  - this happens always for $m < n$, i.e., when the robot is redundant for the task

Mobility analysis for a planar 3R robot

whiteboard …



$l_1 = l_2 = l_3 = 1 \quad n = 3, \quad m = 2$

$WS_1 = \{p \in \mathbb{R}^2 : \|p\| \leq 3\} \subset \mathbb{R}^2$

$WS_2 = \{p \in \mathbb{R}^2 : \|p\| \leq 1\} \subset \mathbb{R}^2$

$$p = \begin{pmatrix} c_1 + c_{12} + c_{123} \\ s_1 + s_{12} + s_{123} \end{pmatrix}$$

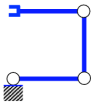
in $\mathbb{R}^2$ in $\mathbb{R}^3$

$$v = \dot{p} = \begin{pmatrix} -s_1 - s_{12} - s_{123} & -s_{12} - s_{123} & -s_{123} \\ c_1 + c_{12} + c_{123} & c_{12} + c_{123} & c_{123} \end{pmatrix} \dot{q} = J(q)\dot{q}$$

case 1)

$q = (0, \pi/2, \pi/2)$

$$J = \begin{pmatrix} -1 & -1 & 0 \\ 0 & -1 & -1 \end{pmatrix}$$



case 2)

$q = (\pi/2, 0, \pi)$

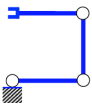
$$J = \begin{pmatrix} -1 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$



- run the MATLAB code subspaces_3Rplanar.m available in the course material

Mobility analysis for a planar 3R robot

whiteboard …



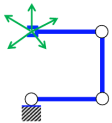
$q = (0, \pi/2, \pi/2)$

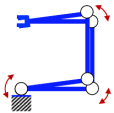
case 1)

$$J = \begin{pmatrix} -1 & -1 & 0 \\ 0 & -1 & -1 \end{pmatrix} \qquad J^T = \begin{pmatrix} -1 & 0 \\ -1 & -1 \\ 0 & -1 \end{pmatrix}$$

$$\rho(J) = 2 = m \qquad \rho(J^T) = \rho(J) = 2$$

$$\mathcal{R}(J) = \text{span}\left\{\begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\right\} = \mathbb{R}^2 \qquad \mathcal{N}(J) = \text{span}\left\{\begin{bmatrix}1\\-1\\1\end{bmatrix}\right\} \qquad \begin{matrix}\dim \mathcal{N}(J) = 1 \\ = n - \rho(J) (= n - m)\end{matrix}$$





$$\mathcal{R}(J) \oplus \mathcal{N}(J^T) = \mathbb{R}^2$$

$$\mathcal{R}(J^T) \oplus \mathcal{N}(J) = \mathbb{R}^3$$

$$\mathcal{R}(J^T) = \text{span}\left\{\begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}0\\1\\1\end{bmatrix}\right\} \qquad \begin{matrix}\dim \mathcal{R}(J^T) = 2 \\ = \rho(J) \ (= m)\end{matrix} \qquad \mathcal{N}(J^T) = 0$$

Mobility analysis for a planar 3R robot

whiteboard …



$q = (\pi/2, 0, \pi)$

case 2)

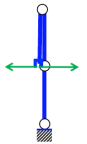
$$J = \begin{pmatrix} -1 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \qquad J^T = \begin{pmatrix} -1 & 0 \\ 0 & 0 \\ 1 & 0 \end{pmatrix}$$

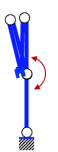
$\rho(J) = 1 < m$ $\qquad \rho(J^T) = \rho(J) = 1$

$\mathcal{R}(J) = \text{span}\left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$ $\qquad \dim \mathcal{R}(J) = 1 = \rho(J)$

$\mathcal{N}(J) = \text{span}\left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$ $\qquad \dim \mathcal{N}(J) = 2 = n - \rho(J)$

forbidden!





$\mathcal{R}(J) \oplus \mathcal{N}(J^T) = \mathbb{R}^2$

$\mathcal{R}(J^T) \oplus \mathcal{N}(J) = \mathbb{R}^3$

$\mathcal{R}(J^T) = \text{span}\left\{ \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$ $\qquad \dim \mathcal{R}(J^T) = 1 = \rho(J)$

$\mathcal{N}(J^T) = \text{span}\left\{ \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}$ $\qquad \dim \mathcal{N}(J^T) = 1 = m - \rho(J)$

Singularities

- **Algebra**

  Jacobian matrix becomes singular

- **Geometry**

  The joint screws (lines) are linearly dependent

- **Kinematics**

  The manipulator (instantaneously) loses one or more degrees of freedom

- **Statics**

  There exists one or more wrenches that can be resisted without turning on the actuators

Kinematic singularities



- **configurations where the Jacobian loses rank**
  $\Leftrightarrow$ **loss of instantaneous mobility** of the robot end-effector
- for $m = n$ (≤ 6), they correspond to Cartesian poses at which the number of solutions of the inverse kinematics problem differs from the generic case
- "in" a singular configuration, we cannot find any joint velocity that realizes a desired end-effector velocity in some directions of the task space
- "close" to a singularity, large joint velocities may be needed to realize even a small velocity of the end-effector in some directions of the task space
- finding and analyzing in advance the mobility of a robot helps in singularity avoidance during trajectory planning and motion control
  - when $m = n$: find the configurations q such that $\det J(q) = 0$
  - when $m < n$: find the configurations q such that all $m \times m$ minors of $J(q)$ are singular (or, equivalently, such that $\det(J(q)J^T(q)) = 0$)
- finding all singular configurations of a robot with a large number of joints, or the actual "distance" from a singularity, is a complex computational task

What is lost in the stretched pose?

$l_1 = l_2 = 1\ m$, $q_1 = q_2 = 0$

$$J = \begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix}, \quad \begin{bmatrix} \dot{x}_P \\ \dot{y}_P \end{bmatrix} = \begin{bmatrix} 0 \\ 2\dot{q}_1 + \dot{q}_2 \end{bmatrix}.$$

$\operatorname{rank} J = 1$.

The image of J is the vertical line.

Command to the right

$\dot{p} = (0.5, 0)$ m/s.

There is no solution for $\dot{q}$.

Command upward

$\dot{p} = (0, 0.5)$ m/s.

$2\dot{q}_1 + \dot{q}_2 = 0.5$.

Many solutions.

A workspace-boundary point may still lack the required *instantaneous* velocity.

Test the 2R singularity in the model

Point position on the left, attainable linear velocities on the right

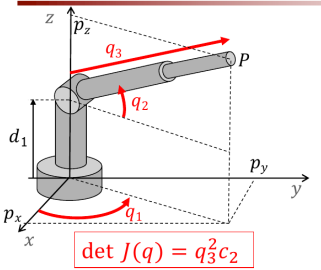
1. Keep the orange arrow pointing right. Change q_2: 60° → 5° → 0°.
2. What happens to the blue velocity region and required $\|\dot{q}\|$?
3. At $q_2 = 0$, turn the arrow upward. Why does the result change?
4. Compare the metrics shown by the model.

Open the interactive model

http://localhost:8765/

The blue region is $\{J\dot{q} : \|\dot{q}\| \leq 1\}$; the orange-arrow tip is the requested velocity $\|\dot{p}\| = 0.5$ m/s.

Singularities of polar (RRP) robot



$$\det J(q) = q_3^2 c_2$$

direct kinematics

$$p_x = q_3 c_2 c_1$$
$$p_y = q_3 c_2 s_1$$
$$p_z = d_1 + q_3 s_2$$

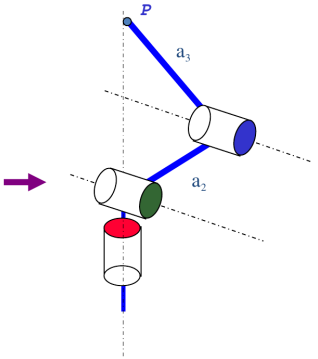
analytic Jacobian

$$\dot{p} = \begin{pmatrix} -q_3 s_1 c_2 & -q_3 c_1 s_2 & c_1 c_2 \\ q_3 c_1 c_2 & -q_3 s_1 s_2 & s_1 c_2 \\ 0 & q_3 c_2 & s_2 \end{pmatrix} \dot{q} = J(q)\dot{q}$$

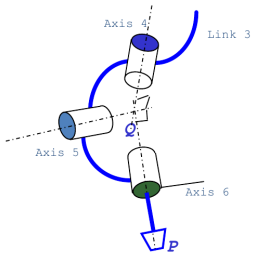
- singularities
  - E-E is along the z axis ($q_2 = \pm\pi/2$): simple singularity ⇒ rank $\rho(J) = 2$
  - third link is fully retracted ($q_3 = 0$): double singularity ⇒ rank $\rho(J)$ drops to 1
- all singular configurations correspond here to Cartesian points internal to the workspace (assuming no range limits for the prismatic joint)

Singularities: Examples

- Case 1: the manipulator is completely extended or flexed
- Case 2: the tool reference point lies on axis 1
- Case 3: orientation singularity
  Axes 4 and 6 are colinear



Singularities: Case 2



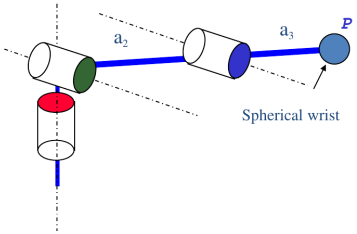
Axes 4 and 6 become colinear

Case 2 $S_5 = 0$

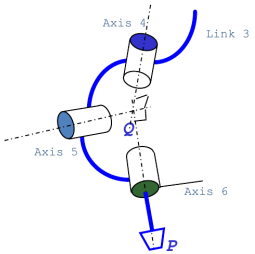
$$\mathbf{J} = \begin{bmatrix} s_{23} & 0 & 0 & 0 & s_4 & 0 \\ 0 & 1 & 1 & 0 & -c_4 & 0 \\ -c_{23} & 0 & 0 & 1 & 0 & -1 \\ \hline -lc_{23} & ms_3 & 0 & 0 & nc_4 & 0 \\ -mc_2 & 0 & 0 & 0 & ns_4 & 0 \\ -ls_{23} & -mc_3 & 0 & 0 & 0 & 0 \end{bmatrix}$$

- Axes 4 and 6 are dependent
- Joints 4 and 6 have the same instantaneous motions
- The end effector loses a degree of freedom

Singularities: Examples



Manipulator is completely flexed/extended



Axes 4 and 6 become colinear

How close is a singularity?

Different metrics answer different questions

Metric	2R formula	Meaning
Pose distance	$\min(\lvert q_2\rvert, \lvert \pi - \lvert q_2\rvert\rvert)$	Angular distance in configuration space
Manipulability	$w = \sqrt{\det(JJ^{\mathsf{T}})} = \prod_i \sigma_i$	Velocity-ellipse area is proportional to w
Weak direction	$\sigma_{\min}(J)$	Smallest velocity stretch
Anisotropy	$1/\kappa_2(J) = \sigma_{\min}/\sigma_{\max}$	Minor-to-major ellipse-axis ratio

Notation. σ_i: singular values of J; $\sigma_{\min}$ and $\sigma_{\max}$: smallest and largest; w: manipulability; κ_2: 2-norm condition number.

Caution

Comparing linear and angular components of a full Jacobian requires an explicit length scale.

Why one determinant is not enough

Mathematical matrices, not models of specific robots

Notation. J_A and J_B are comparison matrices.

$$J_A = \begin{bmatrix} 10 & 0 \\ 0 & 0.1 \end{bmatrix}, \qquad J_B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

	J_A	J_B
$w = \lvert \det J \rvert$	1	1
$\sigma_{\min}$	0.1	1
$1/\kappa_2$	0.01	1
Required $\lVert \dot{q} \rVert$ for $\dot{p} = (0, 1)^{\mathsf{T}}$	10	1

Both matrices are nonsingular; their equal-area ellipses hide different weak directions. For $\dot{p} = (0, 1)^{\mathsf{T}}$, $\dot{q}_A = (0, 10)^{\mathsf{T}}$ and $\dot{q}_B = (0, 1)^{\mathsf{T}}$. Compare $\sigma_{\min}$ and conditioning with consistent units and scales.

What should an engineer do?

Scenario

A robot drives a tool point along a line near a stretched pose. A small linear velocity in the weak direction can demand an infeasible joint speed.

- Change the configuration, approach, tool orientation, or part placement.
- Reduce speed on the problematic segment when needed.
- If a task-space straight line is not required, choose another path after checking limits and collisions.

Exit question

At $q_1 = q_2 = 0$, can a stretched 2R arm move P right? Up? Why?

Reference Materials

- A. De Luca — 11 DifferentialKinematics
- A. Klimchik — L4 Parallel Manipulators M Singularity Analysis

Deserve “A” grade!

– Oleg Bulichev

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